

Pitahaya Aging Diagnostic by Impedance/Capacitance Spectroscopy

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Abstract Impedimetric/capacitive analyses were used as a non-destructive analytical tool for the determination of yellow pitahaya (*Selenicereus megalanthus* Haw.) aging. Pitahaya is an exotic fruit of increasing commercial importance, in which sensing the level of maturation plays a major role. The impedimetric analysis of the fruit's normal deterioration was performed with a homemade probe with two needle silver electrodes. The most sensitive changes on the impedance/capacitance spectra were found at a low-frequency region, where it was possible to build a linear analytical curve between the instrumental response and the aging time.

Keywords Bioimpedimetry · Biosensor · Fruit analysis · Food quality

Introduction

The analysis of the biochemical aspects of fruits and vegetables along with its biological aging are important

analytical parameters for the safety and quality of the food industry. It is the industry's objective to deliver a safe product with a suitable controlled taste aiming for the consumer's satisfaction. Thus, this scenery leads to any new tools of food analysis, particularly, if they are automatic, portable, and non-destructive, to be always welcomed by producers.

A large class of sensorial analysis (including methodologies within the HACCP index) are based on electric variables such as voltage, current, and/or charge, which give origin to robust methods able to be fully automated, rapid, and sensitive (Escarpa 2012). Among the sensorial techniques based on electrical signal for monitoring biological systems, impedance-based technologies such as interdigitated microelectrodes and chip-based impedance microbiology have been increasing (Yang and Bashir 2008; Wu et al. 2008). Impedance spectroscopy (IS) is a very simple and easy technique to be conducted; it has been used to estimate the physiological state of various biological tissues (Jaffrin 2009; Wu et al. 2008), including fruits (Rehman et al. 2011).

Yellow pitahaya is an exotic fruit (Fig. 1) produced in a few parts of the globe; it has had an increased production and commercial success due to its attractive colors and shape as well as its pleasant sensorial properties (Le Bellec et al. 2006; Corrales-García and Canche-Canche 2008; Ortiz-Hernández and Carrillo-Salazar 2012; de Freitas and Mitcham 2013; de Lima et al. 2013). However, it is a seasonal fruit, with two harvesting seasons (in Colombia: February/March and July/August), thus its market growth depends strongly on the aging control, aiming for a safe logistic delivery around the world (de Freitas and Mitcham 2013). In the present work, it is being proposed that the use of impedimetric/

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Fig. 1 Picture of yellow pitahaya (*Selenicereus megalanthus*) taken by Fibonacci, image provided under a GNU Free Documentation license version 1.2



capacitive spectroscopy approach is a simple way of aging control of pitahayas, which may lead to a better assessment of its maturation state and thus possibly develop better conservation conditions that may extend the fruit's commercial time span.

Experimental Procedure

Impedimetric measurements were performed with a HIOKI (Nagano, Japan) 3532-50 equipment and using a homemade probe specially designed for measurements on fruits. The probe consisted of two needle silver electrodes, with a fixed distance between them of 0.7 cm, which is also the value of the needle penetration depth on the fruit sample. AC frequencies ranged from 40 Hz to 5 MHz, with amplitude of 500 mV were used for impedance measurements. All the obtained impedance data were checked regarding compliance with the constraints of linear systems theory by Kramers-Kronig. The modeling and fitting discussed were done using software ZView 2.

Temperature of analysis was kept around 298 K. The fruit samples started to be studied in the third maturation degree (suitable time for fruit harvesting and distribution) (NTC 3554, Frutas frescas. Pitahaya Amarilla 1996).

Fruit samples (*Selenicereus megalanthus* Haw.) were provided by Asopitaya (Valle del Cauca, Colombia).

Results and Discussion

Yellow pitahayas were analyzed in an interval of 8 days, at room temperature and normal atmosphere. They changed from the third to sixth maturation degree established accord-

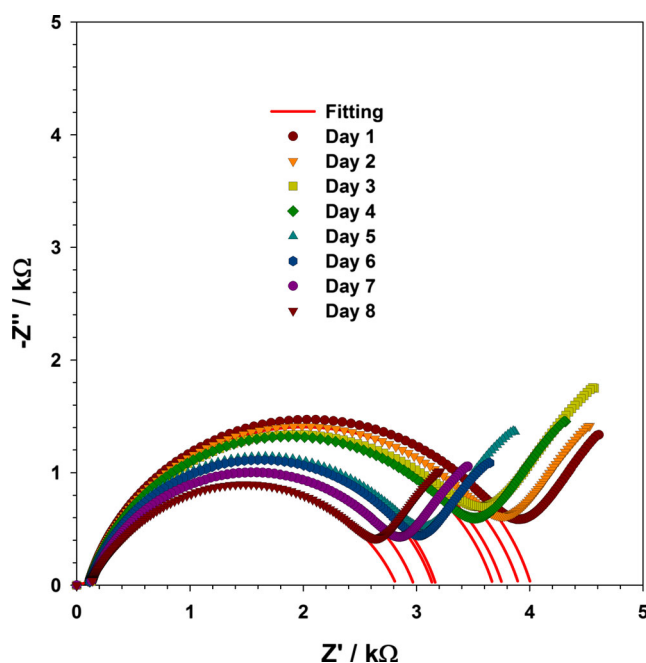


Fig. 2 Impedimetric Nyquist diagram and corresponding fittings of the pitahaya along 8 days of analysis

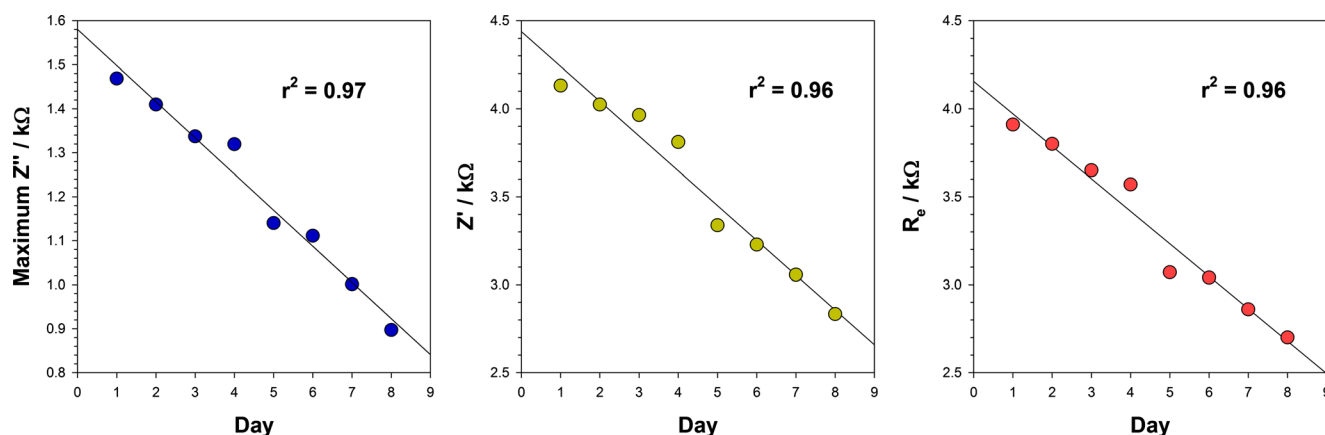


Fig. 3 Linear correlation between maximum Z'' , Z' , and R_e along the 8 days of measurements

ing to an empirical color classification for this fruit that ranks the degree of maturation (NTC 3554, Frutas frescas. Pitahaya Amarilla 1996). The interior maturation was also confirmed by a pH change from 4.9 to 5.3, and also in terms of degrees Brix (which assesses the sugar content of an aqueous solution), from 15.4 to 16.3 Brix.

The impedance response along the days is shown in traditional Nyquist diagram representations (the plot of imaginary Z'' versus real Z' component of the impedance Z^*) in Fig. 2. The Nyquist behavior consists of no ideal semicircle pattern at high frequencies with a straight line variation at higher frequencies, which can be due to an interfacial effect of contact between the fruit tissue and electrodes, i.e., phenomena due to interfaces likely associated to protonic transport according to the natural hydration feature of biological media.

Furthermore, the semicircle region of the spectra was fitted to an equivalent circuit model, broadly used in bioimpedimetric analysis of biological tissues. The total impedance (Z_t) of this equivalent circuit is then given by:

$$Z_t = R_i + \frac{R_e - R_i}{1 + (R_e - R_i)C_t(j\omega)^\alpha}$$

Table 1 Resistive parameter values obtained after the fitting of experimental data to the equivalent circuit mentioned in the text

Days	$R_e/k\Omega$	R_i/Ω	α
1	0.391	102	0.82
2	0.380	105	0.81
3	0.365	107	0.81
4	0.357	111	0.81
5	0.307	111	0.81
6	0.304	112	0.80
7	0.286	118	0.78
8	0.270	124	0.74

where R_i and R_e are the intracellular and extracellular resistance, respectively (Wu et al. 2008). C_t is a non-ideal capacitance associated with the bulk of the fruit sample. ω is the angular frequency and α is a parameter associated with the dispersion of the non-ideal capacitance. The α value is in between the null and unit value and as closer it is from the unit more ideal is the bulk capacitance.

In Fig. 3, it is possible to visualize the linear dependence of R_e , as well as the maximum imaginary component (maximum Z'') and the sole real component obtained at fixed frequency of 2.4 kHz, along the 8 days. Linear behaviors were obtained for all of them, with linearity correlation coefficient higher than 0.95. The linear relationship confirms that the decrease in R_e is likely related with protonic conductance since the mass loss is primarily associated with water content (that also decreases linearly a total of 11 % during the 8 days, data not shown), increasing the protonic concentration so that increasing the fruit tissue conductivity (or decreasing R_e as a consequence) along the aging process. The α values obtained from the performed fitting are around 0.8 and do not vary significantly (less than 8 %) along the days, indicating that the capacitive behavior has around 20 % constant deviation of the ideal value as shown in Table 1.

All these results are very important from a practical point of view since they show the usefulness of the technique to follow the natural fruit aging. More importantly, the visual traditional aging methodology can then be, in a near future, substituted by a quantitative impedimetric analysis.

Conclusions

The obtained data from fitting of the impedimetric data obtained with a homemade probe was shown to be a valuable tool to follow the natural aging of pitahayas. A suitable daily correlation was obtained for the maximum Z'' , Z' , and R_e , thus these parameters, alone or combined, can be used to assess the fruit's aging.

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Conflict of Interest Henry Cortes has no conflict of interest. Villegas Edgar has no conflict of interest. Rócio Suárez has no conflict of interest. José Castillo has no conflict of interest. Luís Moreira Gonçalves has no conflict of interest. Paulo R. Bueno has no conflict of interest. This article does not contain any studies with human or animal subjects.

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